CAMBRIDGE IGCSE & O LEVEL COMPUTER SCIENCE (2210 / 0478)

Topic 2.1: Types and Methods of Data Transmission

COMPLETE TEACHER MARK SCHEME & WORKED SOLUTIONS

Question 1: Data Packets and Packet Structure [Total: 6 Marks]

(a) Three primary parts of a data packet [3 marks]:

- Packet Header [1]
- Payload (or user data / message data) [1]
- Packet Trailer [1]

(b) Three items of information found in a packet header [3 marks — any 3 from]:

- Sender's IP address / Source IP address [1]
- Receiver's IP address / Destination IP address [1]
- Packet sequence number (or packet ID) [1]
- Size of the packet / Packet length [1]

> 💡 *Examiner Tip:* Do NOT accept "checksum" or "CRC" for the header; checksum is located in the trailer.

Question 2: Packet Switching Mechanism [Total: 4 Marks]

1 mark per valid point up to a maximum of 4:

- The original data/file is split into individual numbered packets before transmission. [1]
- Each packet travels independently across the network via routers. [1]
- Different packets may take different routes depending on network traffic and availability. [1]
- Packets can arrive at the destination computer out of order. [1]
- The destination computer uses the sequence numbers (in the header) to reassemble the packets in the correct order. [1]
- The trailer checksum is checked; if any packet is missing or corrupted, a retransmission request is sent. [1]

Question 3: Direction of Data Transmission [Total: 6 Marks]

Scenario	Transmission Mode (1 mark each)	Reason (1 mark each)
Walkie-Talkies	**Half-Duplex** [1]	Data travels in both directions, but only one person can speak at a time. [1]
Computer to printer	**Simplex** [1]	Data travels in one direction only (from computer to printer). [1]
Live video call	**Full-Duplex** [1]	Data travels in both directions simultaneously (both can speak and listen at the same time). [1]

Question 4: Serial vs. Parallel Transmission [Total: 6 Marks]

(a) Table completion [4 marks — 1 mark per correct row]:

Statement	Serial	Parallel
Bits sent one bit at a time down a single wire.	✓ [1]	
More suitable and reliable for long-distance communication.	✓ [1]	
Higher risk of bit skew occurring during transmission.		✓ [1]
Used for internal computer buses over short distances.		✓ [1]

(b) Bit skew definition and problem [2 marks]:

- *Definition:* When bits transmitted at the same time across parallel wires arrive at slightly different times. [1]

• *Consequence:* Data becomes unsynchronized / out of order, leading to corrupted data that cannot be reassembled reliably. **[1]**

Question 5: Universal Serial Bus (USB) **[Total: 4 Marks]**

(a) Benefits of USB [2 marks — any 2 from]:

• Plug-and-play: automatically detected and device drivers configured automatically. **[1]**
• Provides electrical power to peripherals (eliminates external power adapters). **[1]**
• Backward compatible (older USB devices work in newer ports). **[1]**
• Universal industry standard (cannot be connected incorrectly, replaces legacy ports). **[1]**

(b) Drawbacks of USB [2 marks — any 2 from]:

• Cable length is limited (maximum around 5 meters without repeaters/boosters). **[1]**
• Slower transmission speed compared to internal buses (e.g. PCIe). **[1]**
• Older standards (e.g. USB 2.0) have relatively low maximum bandwidth. **[1]**

Question 6: Transmission Application Scenario **[Total: 4 Marks]**

(a) Recommendation and reasons for 8 km sensor link [3 marks]:

• *Recommendation:* **Serial transmission** **[1]**
• *Reason 1:* No risk of bit skew over long distance (8 km). **[1]**
• *Reason 2:* Significantly cheaper cabling and hardware over long distance (fewer wires needed). **[1]**
• *(Alternative reason):* Less electrical interference / crosstalk over long cables. **[1]**

(b) Direction mode required for simultaneous two-way transmission [1 mark]:

• **Full-duplex** **[1]**